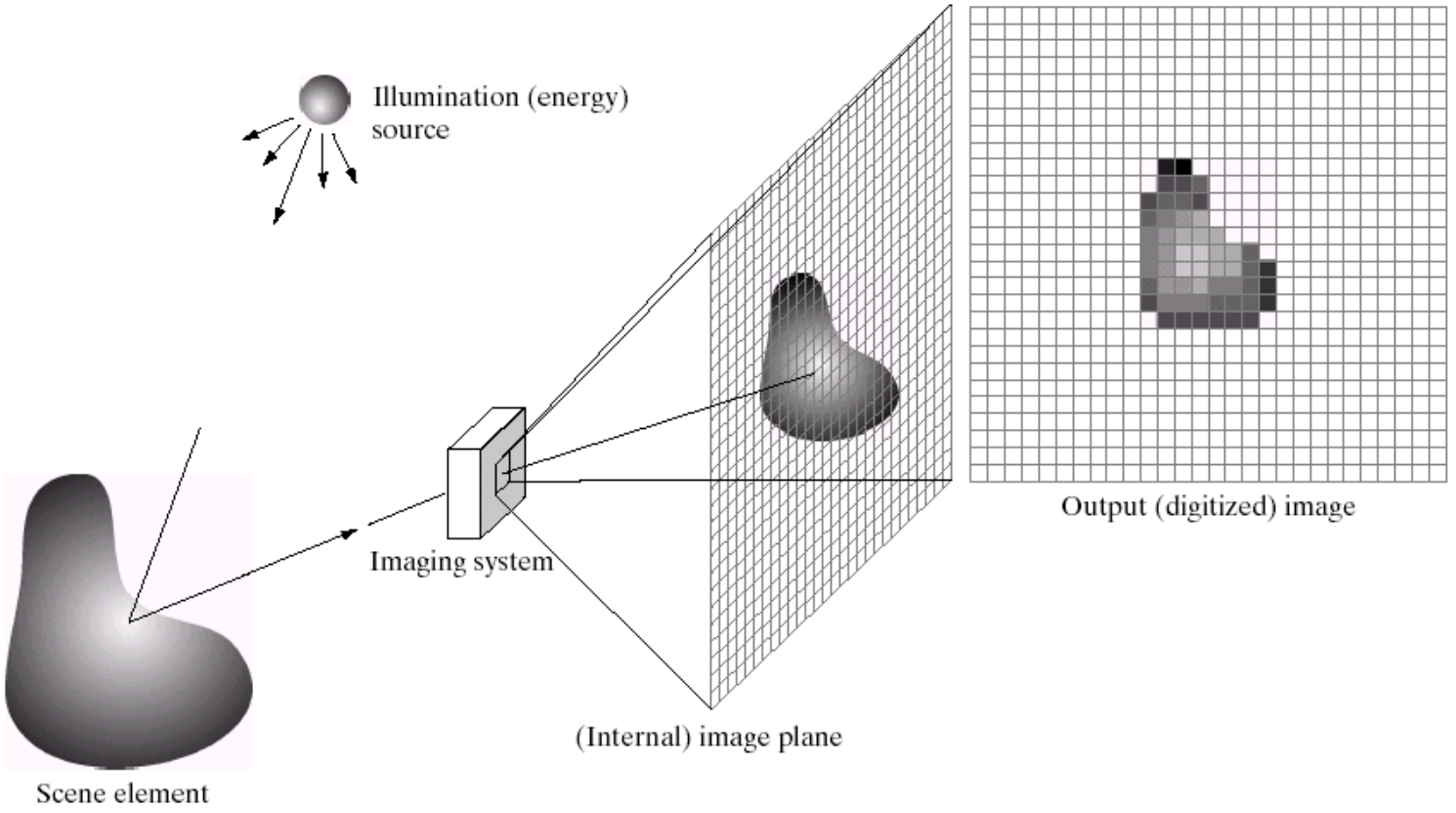

بِسْمِ اللَّهِ الرَّحْمَٰنِ الرَّحِيمِ

Analog Image

- **Images with continuous gray tone or color like a photograph, paintings**
- **What we see in an analog image is various levels of brightness and colors**
- **It is generally continuous and not broken into many small individual pieces (pixels)**

Digital Image

A digital image is a representation of a two-dimensional image as a finite set of digital values, called pixels

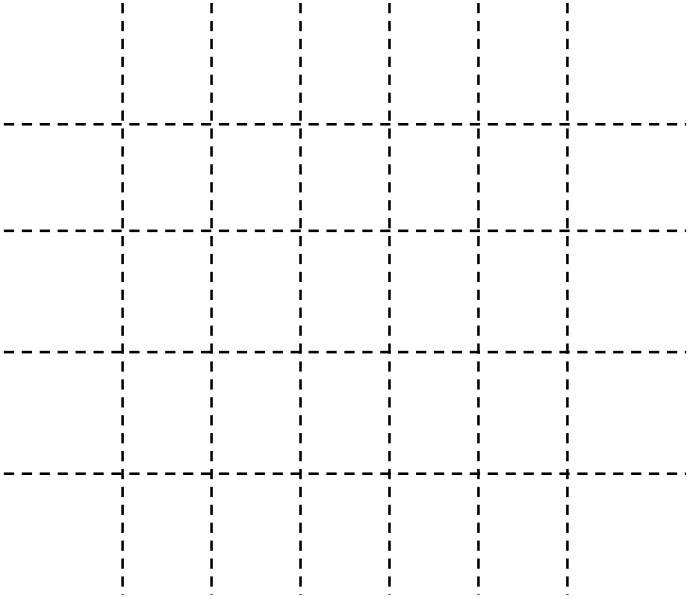


Digital Image

- **Digitization:**
  - **The process of converting the continuous representation in to a discrete ones, discretizing function results as a digital image**
  - **It is obtained by**
    - **Sampling**
    - **Quantizing**
- **Sampling.**
  - **The process discretizing the spatial variables**
  - **The spatial division into a group of cells**

Digital Image

- **Sampling**
  - **Subdividing the image into small equal-sized and shaped areas, called pixels**



Digital Image

- **Quantization**
  - **The continuous radiance of the scene is quantized in to discrete gray levels to represent the scene radiance for each pixel**
  - Representing the brightness of each area with a numeric value or **digital number**

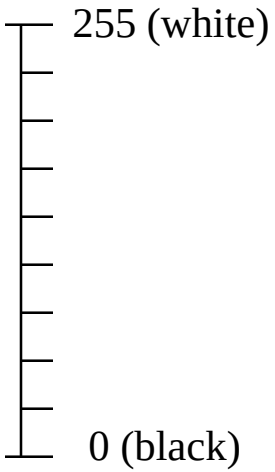


Photo Representation in Digital Format

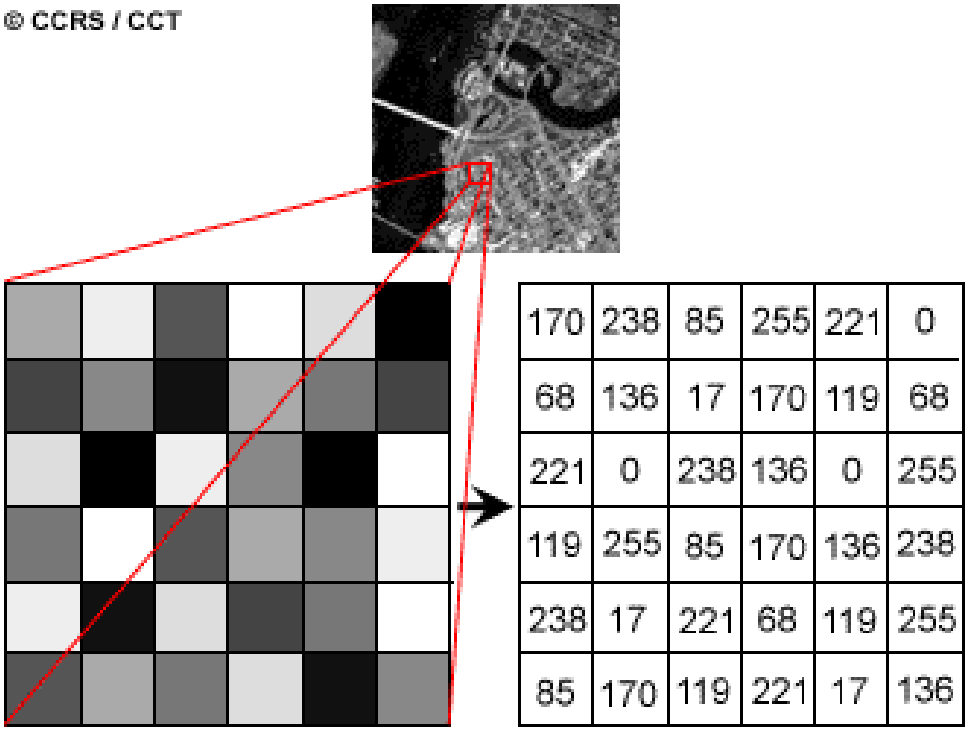


Image Representation-Notation

- **The number of pixels in the image are denoted with n,**
- **Spatial position of the pixel at row r and column c is denoted (r,c)or [i,j]**
- **Another convention used is to use x for c and y for r, (x,y)**

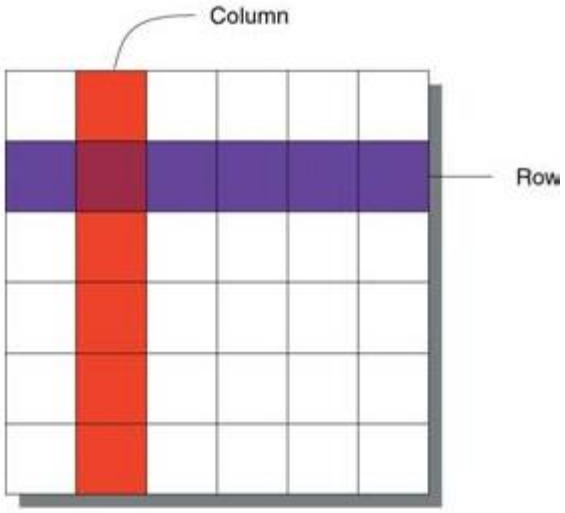
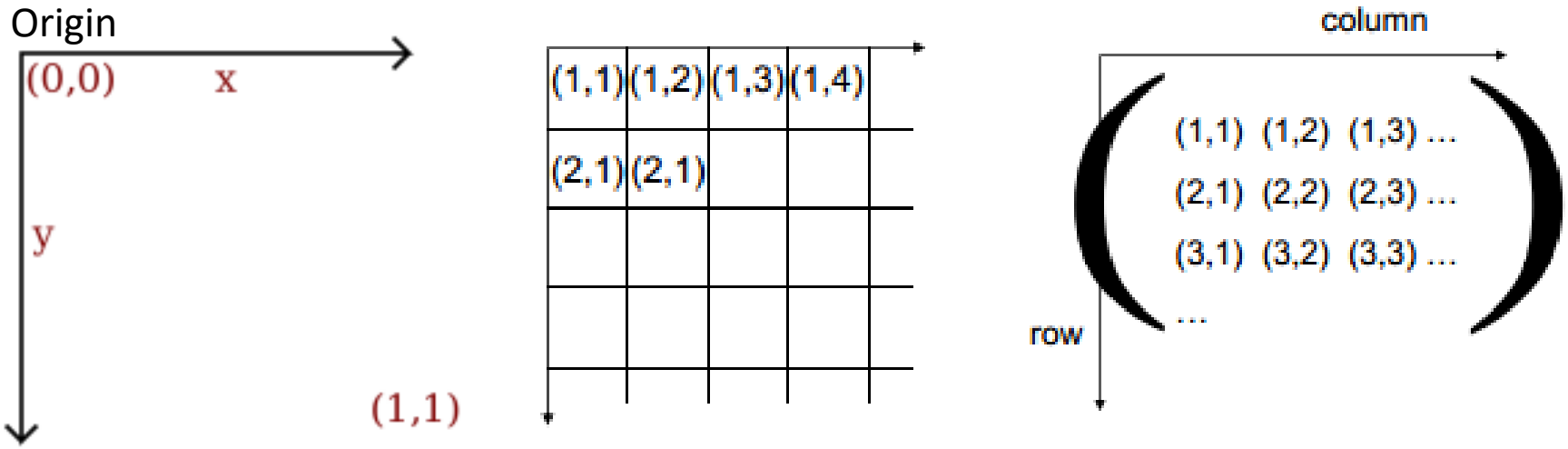


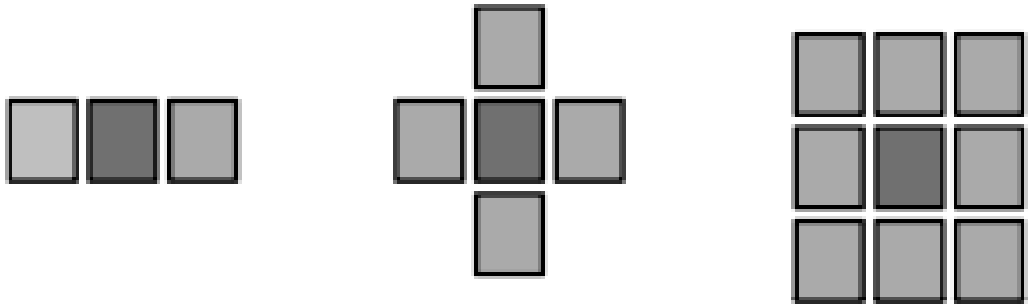
Image Representation



In the digital images, the row , col start from upper left corner of the image

Image Pixel's Neighborhood

- Neighborhood: 2+1 pixels,4+1 pixels,...
- 3x3 neighborhood: 8+1 pixels



Remote Sensing Metadata

- ***Metadata* is "*data or information about data*"**
- **As an image file contains only pixel values, additional information is needed in order to display it, and to keep track of it**
- **Depending on the software used, this metadata is located either at a separate file (usually an ASCII file with the same name, using a different extension), or in the same file containing the data, before the data**
- **This information is known as *header, or documentation.***

Remote Sensing Metadata

- Metadata might include
  - The file name
  - Date of last modification
  - Level of quantization (e.g., 8-bit)
  - Number of rows and columns
  - Number of bands
  - Univariate statistics (minimum, maximum, mean, median, mode, standard deviation), may be some multivariate statistics
  - Geo-referencing performed (if any)
  - Pixel size

Auxiliary Data

- **An image scene is composed of multiple files, each of which is composed of multiple records.**
- **Data other than image data in the files, are called Auxiliary Data**
- **Auxiliary Data is supplementary information used to complement, validate, or enhance the analysis of a primary dataset**

